

Total No. of Questions : 10]

SEAT No. :

PA-341

[Total No. of Pages : 3

[59271-225

B.E. (Mechanical/Mechanical(Sandwich))

ENERGY ENGINEERING

(2015 Pattern) (Semester - II) (402047)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right of each question indicate full marks.
- 4) Assume suitable data wherever necessary and mention the same clearly.
- 5) Use of steam tables, mollier chart and calculator is allowed.

Q1) a) Elaborate following terms in brief : **[5]**

- i) Load Shedding.
- ii) Carbon Credits.

b) Define following terms : **[5]**

- i) Vacuum Efficiency.
- ii) Condenser Efficiency.
- iii) Corrected Vacuum.

OR

Q2) a) Which are factors considered for selection of coal for Thermal Power Plant? **[5]**

b) What do you mean by FGR? Explain with simple diagram. **[5]**

Q3) a) Elaborate working principal of FBC. What are different types of FBC? **[5]**

b) What is a function of Surge tank in Hydropower Plant? Enlist different types of the same and explain anyone from it. **[5]**

OR

P.T.O.

Q4) a) Explain typical construction, working and advantages of PWR with simple diagram. [5]

b) What do you mean by ESP? Explain with simple diagram. [5]

Q5) a) Explain General Layout of Diesel Power Plant indicating different systems. [8]

b) The pressure ratio used in an open cycle gas turbine power plant is 6.5. The pressure and temperature of air entering in the compressor are 1 bar, 300K. Inter cooling arrangement is used to reduce the work of compression. The maximum temperature of the cycle is limited to 850K. If the power plant capacity is 10MW, find the thermal efficiency of the plant and air consumption per hour if calorific value is 45000kJ/kg. Assume compression in both the stages and expansions in turbine are isentropic. Take $\gamma = 1.4$ (air and gas), $C_p = 1 \text{ kJ/kgK}$ (air and gas). Assume perfect inter cooling. [8]

OR

Q6) a) Explain the advantages, disadvantages and applications of diesel power plant in brief. [8]

b) An open cycle gas turbine plant works on Brayton cycle. The maximum pressure and temperature of the cycle is limited to 5 bar, 900K. The pressure and temperature at the inlet to the compressor are 1 bar and 300K. Reheating is used at the pressure of 2.5 bar where the temperature of the gases is increased to its original turbine inlet temperature. Mass flow rate of the air in the plant is 10 kg/s. Determine the thermal efficiency and capacity of the plant in MW. The exhaust pressure of the gases leaving the plant is 1 bar. Assume compression and expansions are isentropic. Take $\gamma = 1.4$ (air and gas), $C_p = 1 \text{ kJ/kgK}$ (air and gas), CV of the fuel 40000 kJ/kg. Neglect pressure losses in the system. Do not neglect the mass of the mass of the fuel. [8]

Q7) a) Explain Low temperature flat plate collector solar power plant with a suitable sketch. [6]

b) Explain Superheated Geothermal Power Plant a suitable diagram. [6]

c) What is working principal of Solar Photovoltaic System? What are advantages and applications of the same. [6]

OR

- Q8)** a) Explain Open cycle OTEC with typical diagram and merits. [6]
 b) Explain Solar Chimney with typical diagram and merits. [6]
 c) Elaborate advantages and disadvantages of MHD based plant. [6]

- Q9)** a) What is the function of switch gear? Explain any one type of switchgear with neat diagram. Discuss their relative merits and demerits. [8]
 b) A power station has to supply load as follows : [8]

Time (hours)	0-6	6-12	12-14	14-18	18-24
Load (MW)	30	90	60	100	50

- Draw
- i) Load Curve.
 - ii) Load duration curve and Calculate.
 - 1) Load factor
 - 2) Capacity of the plant and
 - 3) Plant capacity factor.

OR

- Q10)** a) Explain terms with significance : [8]
 i) Load factor,
 ii) Peak and base load,
 iii) Demand factor,
 iv) Plant capacity factor.
 b) Write note on : [8]
 i) Layout of electrical equipment.
 ii) Methods of generator cooling.

